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No.

ODD END SEMESTER EXAMINATION DECEMBER - 2025

Name of the Course: B.Tech Aerospace Engineering
 Name of the Paper: Introduction to Aerospace Engineering

Semester: III
 Paper Code: TAS 302

Time: 3 Hours

Maximum Marks: 100

Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20 marks)	CO1
(a)	Analyze the configuration and key design features of the Wright Flyer I, developed by the Wright Brothers. How do these elements contribute to its success in powered flight?	
(b)	Evaluate the contributions of Sir George Cayley to the development of aviation. In what ways did his work influence subsequent advancements in aeronautics?	
(c)	Examine the design and configuration of the first successful glider created by Otto Lilienthal. What were the key innovations that led to its success in flight?	
Q2	(20 marks)	CO2
(a)	Develop equations for the pressure and density as a function of altitude from 0-21 KM. The analytical model of the atmosphere should make use of the hydrostatic equation, for which the density is eliminated through the use of the equation of state for a thermally perfect gas. Assume that the temperature of the air from 0-11 KM is given by $T = T_0 - \lambda h$	
(b)	If the pressure and temperature at sea level are standards, calculate the pressure and density at 8000m. Is the lapse rate is (i) 0.0065 °K/m ii) 0.0095 °K/m	
(c)	The altimeter on a low-speed airplane reads 2 km. The airspeed indicator reads 50 m/s. If the outside air temperature is 280 K, what is the true velocity of the airplane?	
Q3	(20 marks)	CO3
(a)	Identify the various types of aerodynamic drag encountered by an aircraft, explain the physical mechanisms responsible for each, and discuss the methods used to minimize them.	
(b)	Classify the various categories of flying vehicles and present the classification in the form of an appropriate flowchart.	
(c)	Define the term circulation. What is the effect of circulation for a flow around a circular cylinder.	

Q4	(20 marks)	
(a)	Construct a neat schematic illustrating the pressure distribution and the corresponding C_p vs. x/c plot for a symmetrical airfoil at angles of attack -5° , -2° , $+2^\circ$, 8° , 15° , and 20° , and analyze the variations observed across these conditions.	CO4
(b)	Explain the significance of structural weight in aircraft design. Additionally, define the following structural design concepts: (a) fail-safe versus safe-life design, (b) limit load, and (c) ultimate (design) load.	
(c)	Describe and compare the various types of fuselage construction. Using neat sketches, illustrate each construction method and explain its structural characteristics.	
Q5	(20 marks)	
(a)	Describe the principal components of a jet engine with neat sketch and analyze the underlying mechanism through which thrust is generated in a gas-turbine propulsion system	CO5
(b)	Differentiate between: (a) turbofan and turboprop engines, and (b) turbojet and turbofan engines, highlighting key operational and performance distinctions	
(c)	Evaluate the advantages and limitations of solid propellant and liquid propellant systems used in rocket propulsion	